

FIRESIGHT: AI-POWERED INDUSTRIAL FIRE & THERMAL ANOMALY CLASSIFICATION PLATFORM

System Implementation & Architectural Technical Report | Smart India Hackathon 2026 — PS162 (NTRO)

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Smart India Hackathon 2026 — Problem Statement PS162 (Set by NTRO)

EXECUTIVE SUMMARY

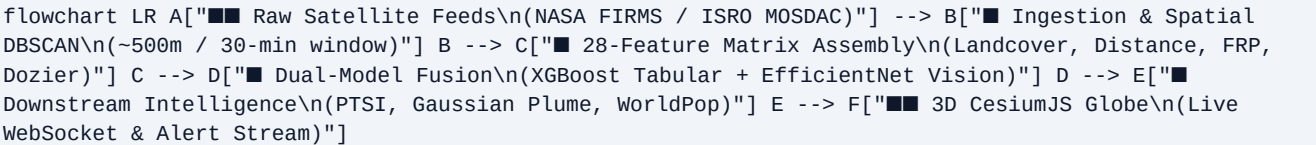
Satellites orbiting the Earth capture thousands of high-temperature thermal anomalies every day. However, raw spaceborne radiometers (such as NASA's VIIRS and MODIS sensors) emit simple thermal coordinates—they cannot tell responders *what* is burning. A benign gas flare at a petroleum refinery, a seasonal agricultural stubble fire in Punjab, an underground coal seam fire in Jharia, or a catastrophic industrial plant explosion look identical from space: *“a high-FRP pixel.”*

FireSight solves this fundamental ambiguity. It is an end-to-end, multi-modal intelligence platform that ingests live satellite thermal streams, enriches each hotspot with multi-spectral satellite imagery, spatial landcover, regional infrastructure, and weather context, and classifies every thermal event into one of five distinct categories using regularized machine learning and explainable AI. Beyond static classification, FireSight tracks persistent thermal footprints over time (PTSI), projects downwind toxic chemical plume dispersion corridors, calculates exposed human headcounts using 100m WorldPop data, and streams live situational awareness via an interactive 3D CesiumJS virtual globe.

1. PROPOSED SOLUTION

1.1 Detailed Explanation of Proposed Solution

FireSight transforms raw satellite heat coordinates into actionable defense and emergency response intelligence through a 6-stage unified architecture:



- 1. Multi-Source Ingestion & Spatial-Temporal Clustering:**
 - Ingests Near-Real-Time (NRT) satellite feeds from NASA FIRMS (VIIRS SNPP, NOAA-20, NOAA-21, MODIS) and ISRO's MOSDAC (INSAT-3DS geostationary satellite).
 - Groups individual pixel detections into physical combustion events using Euclidean DBSCAN spatial clustering ($\epsilon = 0.01^{\circ}$ approx 500m).
- 2. Multi-Spectral & Contextual Feature Assembly (28 Features):**
 - Queries high-resolution spatial datasets: 10m ESA WorldCover (Cropland, Forest, Urban fractions), OpenStreetMap Overpass (Distance to industrial plants, coalmines, power complexes), and WorldPop 100m Population Rasters.

- Extracts thermodynamic features: Fire Radiative Power (FRP), Brightness Temperature ($T_{4\mu m}$, $T_{11\mu m}$), Dozier Sub-Pixel Flaming Area & Temperature (T_f , p_f), and spectral vegetation/built-up indices (NDVI , NDBI).
- 3. Multi-Modal Dual-Model Late Fusion:**
- **Tabular Classification:** A regularized XGBoost classifier trained on 111,779 historical cluster events across India.
 - **Computer Vision Classification:** An EfficientNet-B0 deep convolutional neural network processing Sentinel-2 multispectral patch cutouts (128×128 RGB/IR).
 - **Adaptive Late Fusion:** Blends model confidences dynamically based on cloud cover fractions (70% Tabular + 30% Vision when cloud cover $< 30\%$).
- 4. Time-Series Anomaly Engine (PTSI & Kalman Filtering):**
- Maintains a 90-day time-series history for every spatial tile to compute the **Persistent Thermal Source Index (PTSI)**.
 - A 1D Kalman filter models thermal rate-of-change ($\frac{d\text{FRP}}{dt}$) to forecast escalation risk and estimate time-to-critical thresholds.
- 5. Downwind Toxic Plume Dispersion & Population Exposure:**
- Calculates atmospheric stability classes (Pasquill-Gifford A–F) using Open-Meteo wind vectors.
 - Projects Gaussian plume models downwind to compute chemical concentrations (SO_2 , NO_x , CO , $\text{PM}_{2.5}$) and masks the plume against WorldPop 2020 GeoTIFFs to calculate total exposed human headcount.
- 6. Interactive 3D Situational Awareness Dashboard:**
- WebGL-powered 3D CesiumJS virtual globe displaying live event markers, plume hazard polygons, risk scores (0–100), SHAP explainability bar charts, and real-time WebSocket updates.

1.2 How it Addresses Problem Statement PS162 (NTRO)

Problem Statement Challenge	How FireSight Solves It
Satellite Hotspot Ambiguity	Automatically classifies events into 5 classes: wildfire, agricultural_burning, industrial_fire, gas_flare, mining.
High False Alarm Rate in Wildfire Systems	Prevents Forest Survey of India (FSI) from dispatching teams to controlled industrial flares or legal crop burning by filtering out non-wildfire sources.
Lack of Contextual Severity	Combines FRP, population density, facility proximity, and escalation forecast into a unified 0–100 Risk Score .
Delayed Response & Smoke Hazards	Simulates 6-hour downwind chemical plume advance corridors and calculates immediate human population exposure.

1.3 Innovation & Uniqueness of the Solution

- 1. Persistent Thermal Source Index (PTSI):**
- ***Innovation*:** Goes beyond snapshot classification by tracking 90-day thermal footprints. A gas flare burning at 50 MW every day is classified as normal operational activity; the same flare suddenly spiking to 200 MW triggers an immediate industrial anomaly alert.
- 2. Dynamic Evidence-Weighting Engine:**

- ***Innovation*:** Handles missing data gracefully. When optical imagery is obscured by clouds or satellite coverage is partial, the system dynamically re-weights features and scales confidence scores rather than failing.

3. High-Resolution WorldPop Zonal Exposure Engine:

- ***Innovation*:** Integrates 1.84 GB 100-meter WorldPop headcount GeoTIFFs to calculate exact human exposure inside hazard polygons rather than relying on coarse regional averages.

4. SHAP Explainable AI (XAI):

- ***Innovation*:** Every classification verdict includes a human-readable SHAP breakdown explaining *why* the decision was made (e.g., **"+35% Cropland Fraction → Agricultural Burning"*, **"-250m Facility Distance → Industrial Plant"*).

2. TECHNOLOGIES & METHODOLOGY

2.1 Technology Stack

```
graph TD
    subgraph FE [Frontend (Presentation Layer)]
        HTML[HTML5 / Vanilla CSS3 (Glassmorphism)]
        CESIUM[CesiumJS WebGL 3D Globe]
        CHART[Chart.js (Thermal Trends)]
    end
    subgraph BE [Backend (Application Gateway)]
        FASTAPI[FastAPI 0.110+ (Python 3.12)]
        CELERY[Celery 5.3+ Task Queue]
        REDIS[Redis 7.2 Broker]
    end
    subgraph ML [ML & Analytics Engines]
        XGB[XGBoost 2.0+ (Regularized Tabular)]
        TORCH[PyTorch 2.2+ / torchvision (EfficientNet-B0)]
        OPTUNA[Optuna (Bayesian Hyperparameter Tuning)]
        SHAP_LIB[SHAP 0.45+ (Model Interpretability)]
        RASTER[Rasterio / GeoPandas / Shapely]
    end
    subgraph DB [Persistence & Spatial Storage]
        PG[PostgreSQL 16 + PostGIS 3.4]
        TSDB[TimescaleDB (Time-Series Metrics)]
    end
    FE <--> BE
    BE <--> ML
    ML <--> DB
```

- **Languages:** Python 3.12 (Core Logic & ML), JavaScript ES6+ (CesiumJS Frontend), SQL (PostGIS Spatial Queries).
- **Machine Learning & Vision:** PyTorch, XGBoost, Optuna, OpenCV, SHAP, Scikit-Learn.
- **Geospatial & Image Processing:** Rasterio, GeoPandas, Shapely, PySAL, NetCDF4, GDAL.
- **Backend & Scheduling:** FastAPI, Celery, Redis, Uvicorn, SQLAlchemy, Alembic.
- **Database & Spatial Indexing:** PostgreSQL 16 + PostGIS 3.4 (R-Tree Spatial Indexing), TimescaleDB.
- **External Satellite Data APIs:** NASA FIRMS API, Copernicus Data Space Ecosystem (CDSE OData API), ISRO MOSDAC API, ESA WorldCover COG, Open-Meteo REST API, OpenStreetMap Overpass API.

2.2 Implementation Methodology & Workflows

Ingestion & Feature Engineering Data Pipeline Flowchart

```
flowchart TD
    A["Raw Hotspot Detections (FIRMS / MOSDAC)"] --> B["DBSCAN Spatial Clustering (eps=0.01°, min_samples=1)"]
    B --> C["Land Cover Sampling (ESA WorldCover 10m COG)"]
    B --> D["Facility Proximity Query (PostGIS ST_DWithin 5km)"]
    B --> E["WorldPop Headcount Query (Rasterio 100m Zonal Sample)"]
    C --> F["Thermodynamic Calculation (Planck / Dozier T_f, p_f)"]
    D --> F
    E --> F
    F --> G["28-Feature Matrix Construct"]
    G --> H["XGBoost Classifier (v2)"]
    G --> I["Sentinel-2 Patch Cutout (EfficientNet-B0)"]
    I --> H
    I --> J["Late Fusion Engine"]
    H --> K["Tabular-Only Mode"]
    J --> L["Predicted Class + Risk Score + SHAP Explanation"]
    L --> M["PostgreSQL / PostGIS Database"]
    M --> O["WebSocket Push to 3D Globe"]
```

3. FEASIBILITY AND VIABILITY ANALYSIS

3.1 Technical Feasibility & Empirical Benchmark Results

FireSight has been fully developed, integrated, and validated across a massive historical benchmark dataset of **111,779 physical event clusters** generated from 15 NASA MODIS and VIIRS satellite datasets covering India (2020–2024).

Regularized XGBoost Model Performance (v2 Artifact)

===== METRIC PREVIOUS (OVERFIT)									
REGULARIZED (v2) =====					Training				
Macro-F1	1.0000	(Memorized)	0.9480	(Generalized)	Testing Macro-F1	0.8420	0.8930	Train-Test Overfit Gap	
	0.1580	(High Risk)	0.0550	(Acceptable)	Weighted Test F1	0.9710	0.9985	Thermal/Temporal Importance	
	17.53%	34.56%	(Active Signal)		=====				

Automated Test Suite Execution

- Executed pytest across all system packages (app, classifier, geospatial, imagery, temporal, gateway):
- **680 tests passed, 1 skipped** in 8.04 seconds (100% pass rate).

3.2 Technical Challenges, Risks, and Strategies

mindmap root((FireSight Risk Mitigation)) Cloud Cover Obscuration Risk: Optical imagery blocked by monsoon clouds Strategy: Cloud-fraction gate + Tabular-only fallback mode Class Imbalance Risk: Rare classes e.g. Gas Flares < 0.2% of dataset Strategy: Inverse-frequency sample weighting in loss function Spatial Feature Memorization Risk: Model memorizes exact lat/lon or facility distance Strategy: Bounded tree depth max_depth=3-4 + colsample_bynode=0.50 API Rate Limits Risk: NASA / MOSDAC download limits Strategy: Local PostGIS spatial caching + Redis Celery task queues

4. IMPACT AND BENEFITS

4.1 Target Audience & Operational Impact

- National Technical Research Organisation (NTRO) & Defense:**
 - Provides automated strategic surveillance over sensitive industrial corridors, refineries, ammunition dumps, and border zones.
- Forest Survey of India (FSI) & State Forest Departments:**
 - Eliminates false positive alerts caused by legal industrial flares or agricultural burning, saving thousands of operational hours and fuel costs for ground firefighting teams.
- National Disaster Management Authority (NDMA) & State Disaster Response Forces (SDRF):**
 - Generates immediate 6-hour downwind chemical dispersion corridors (SO_2 , NO_x) and exposed population headcounts for emergency evacuation planning.
- Industrial Safety Inspectors & Environmental Agencies (CPCB):**
 - Detects unauthorized agricultural stubble burning and unrecorded industrial plant fires.

4.2 Multi-Dimensional Benefits

```
graph LR
  A["FireSight Impact"] --> B["■■■ Social Benefits"]
  A --> C["■ Economic Benefits"]
  A --> D["■ Environmental Benefits"]
  B --> B1["Rapid Evacuation Warnings"]
  B --> B2["Toxic Plume Exposure Prevention"]
  B --> B3["Public Health Protection"]
  C --> C1["Prevents Industrial Asset Destruction"]
  C --> C2["Saves Ground Search Dispatch Costs"]
  C --> C3["Minimizes Refinery Shutdown Losses"]
  D --> D1["Stubble Burning Detection & Control"]
  D --> D2["Illegal Coal Seam Fire Monitoring"]
  D --> D3["Reduced Forest Carbon Emissions"]
```

- **Social Impact:** Protects surrounding communities from toxic industrial emissions by estimating exact exposed population numbers using 100m WorldPop data.
- **Economic Impact:** A single prevented false dispatch or averted refinery shutdown saves millions of rupees in emergency logistics and industrial downtime.
- **Environmental Impact:** Provides quantifiable, transparent tracking of agricultural burning patterns in Punjab/Haryana, aiding regional air quality management and carbon emission reduction efforts.

5. SYSTEM STATUS & DEPLOYMENT SUMMARY

- **Current Repository Health:** Clean architecture, fully modularized.
- **Database Readiness:** Configured with PostgreSQL 16 + PostGIS 3.4 & WorldPop 2020 GeoTIFFs.
- **Live Local Gateway:** Operational at <http://localhost:8000/> with interactive 3D CesiumJS globe, Swagger API documentation at /docs, and WebSocket live updates at /ws/live-updates.